

Smart Metro Ticket Booking System

Requirement Analysis Phase

Date	
Team ID	
Project Name	Smart Metro Ticket Booking System
Maximum Marks	4 Marks

3. Solution Requirements

This document brings together the functional and non-functional requirements for the metro ticket booking digitization solution on ServiceNow, along with the component mapping, actor roles, assumptions, and delivery milestones agreed for this phase.

3.1 Functional Requirements

Ticket Booking (Service Catalog)

- A Service Catalog item for ticket booking with variables for Source Station, Destination Station, Passenger Type, and Number of Tickets.
- Station selections are validated against the Station Master to block invalid routes.
- Bookings can be made for multiple passenger types: General, Student, Senior Citizen.
- Multiple tickets can be booked in a single transaction.

Fare Calculation

- Fare is calculated automatically based on the distance/zone between source and destination.
- Passenger-type discounts and multi-ticket totals are applied automatically.
- An itemized fare breakdown is shown before payment is confirmed.

Payment Processing

- Digital payment modes supported: UPI, debit/credit card, and mobile wallets.
- Payment status is confirmed back to the booking flow in real time.
- Every transaction is logged with a unique reference number for audit and reconciliation.

QR Ticket Generation

- A unique QR-coded digital ticket is generated immediately after payment succeeds.
- Each QR code is single-use and valid only for the booked journey window.
- The generated ticket is stored against the passenger's booking record for later retrieval.

Notifications

- The digital ticket and QR code are sent via in-app notification, email, and WhatsApp/SMS.
- Passengers are notified of payment failures along with a retry link.
- Metro Operations is alerted to payment exceptions or QR generation failures.

Administration & Reporting

- Station Managers maintain station and fare master data through a controlled workflow.
- Metro Operations gets dashboards covering ticket volume, revenue, and peak usage.
- IT Admins can configure catalog variables, ACLs, and integration endpoints.

End-to-End Solution Process Flow

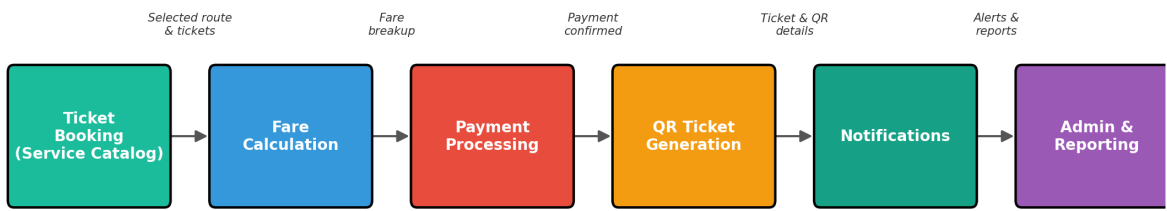


Figure — End-to-end flow across the functional requirement areas above

3.2 Non-Functional Requirements

Category	Requirement
Performance	Fare calculation and QR generation must finish within 3 seconds of payment confirmation under normal load.
Scalability	The solution must handle concurrent booking spikes during peak commute hours without degrading.
Availability	Core booking and QR validation services should target 99.5% uptime during metro operating hours.
Security	All payment data must go through PCI-compliant gateways; role-based access control (ACLs) must govern every sensitive table.
Usability	A first-time user should be able to complete the booking flow in under 5 steps, with accessible, mobile-friendly UI Pages.
Auditability	Every booking, payment, and QR generation event must be logged with a timestamp and correlation ID for traceability.
Maintainability	IT Admins must be able to configure fare rules and station data without needing a code deployment.

3.3 ServiceNow Component Mapping

Requirement Area	ServiceNow Component	Purpose
Ticket Booking Form	Service Catalog + Catalog Client Scripts	Captures Source, Destination, Passenger Type, and Number of Tickets with real-time validation.
Fare Calculation	Business Rules / Script Includes	Runs fare logic server-side based on station distance and passenger type.
Booking-to-Ticket Workflow	Flow Designer	Orchestrates the full flow: payment trigger, QR generation, and notification dispatch.
Payment Integration	Integration Hub / REST Spokes	Connects to external payment gateway APIs for UPI, card, and wallet transactions.
QR Code Generation	Integration Hub / Script Include (QR library)	Generates a unique QR image and links it to the ticket record.
Alerts & Ticket Delivery	Notifications (Email/SMS/Push)	Delivers digital tickets and status alerts to passengers and staff.

Requirement Area	ServiceNow Component	Purpose
Passenger Self-Service	Service Portal / UI Pages	Hosts the booking form, fare preview, booking history, and QR ticket display.
Reporting	Performance Analytics / Reports	Gives Metro Operations usage, revenue, and peak-time dashboards.

ServiceNow Component Architecture

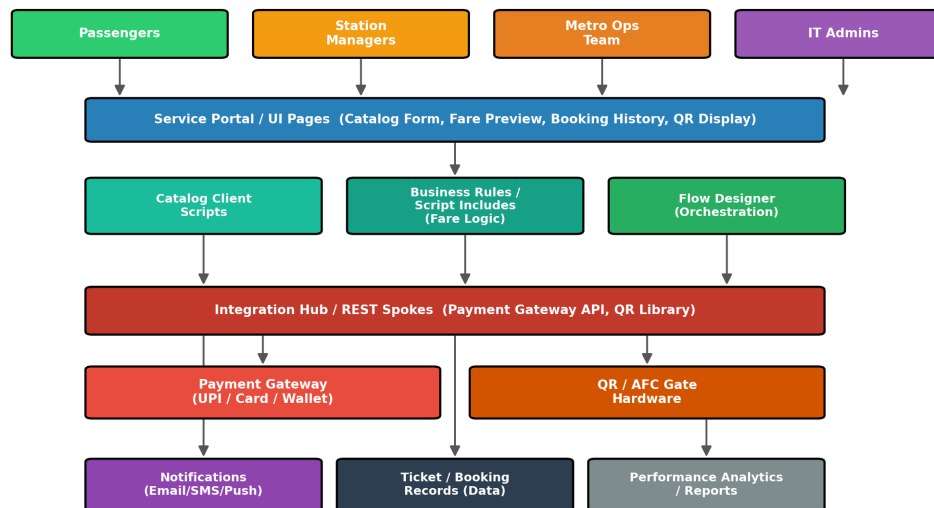


Figure — Layered ServiceNow architecture realizing the component mapping above

3.4 Actors & Roles

Actor	Role in the System
Passengers (End Users)	Book tickets, make payments, receive and use QR tickets, view booking history.
Station Managers	Maintain station/fare master data, validate QR tickets at gates, monitor station-level ticket volume.
Metro Operations Team	Consume analytics and reports, manage exceptions, plan schedules based on demand data.
IT Admins	Configure catalog items, flows, integrations, ACLs, and troubleshoot platform issues.

3.5 Delivery Milestones

#	Milestone	Description
1	Catalog Form Setup	Build the Service Catalog item with Source, Destination, Passenger Type, and Number of Tickets variables.
2	Fare Logic Script	Implement server-side fare calculation logic covering distance/zone and passenger-type rules.
3	QR Code Integration	Integrate the QR generation library/API and link its output to the ticket record.
4	Station Details Mapping	Load and map the Station Master data used for route validation and fare lookup.
5	UI Pages	Design and build the passenger-facing Service Portal pages for booking, fare preview, and ticket display.
6	UAT & Deployment	Carry out user acceptance testing with all personas and deploy to production.

3.6 Assumptions & Constraints

- Passengers have access to a smartphone or a WhatsApp-enabled device to receive digital tickets.
- A payment gateway supporting UPI, card, and wallet is available for integration.
- Station and fare master data will be supplied and kept current by the Metro Operations Team.
- QR-based Automated Fare Collection (AFC) gate hardware is available, or planned, at stations for scanning.